ARALert: Pilot Testing & ISO/IEC 25010

Pre-Deployment, Physical Papers & Evaluation Master Plan

Comprehensive Action Guide & Guide Screenshot Map for 1-Week Pilot Deployment

Target System: ARALert (Web & Offline-Ready Academic Tracking) **Course: IT 124 Capstone Project 2**
Pilot Duration: 1-Week Operational Deployment at Partner School Term: First Semester, SY 2026-2027
Post-Pilot Goal: ISO/IEC 25010 System Evaluation (3 Stakeholder Groups) Authority: Prof. Felix R. Penetrante Guide

1. Executive Overview & Essential Directives

According to the BUCS CSIT Capstone Project 2 Student Guide (IT 124), successful completion of Chapter 4 (Results and Discussion) and a defensible final thesis defense rest entirely on two foundations: empirical evidence gathered from actual deployment, and strict compliance with declared research standards. Chapter 4 is not about intentions?it is the factual record of what happened during your 1-week pilot test and how the system scored under ISO/IEC 25010 user evaluation.

Key Ground Rules from the Guide (Pages 5, 27, 32, 40):
? Complete System Precondition: Section 4.3 reports a finished, tested system?never an unfinished prototype.
? Independent Evaluation: Project developers are STRICTLY FORBIDDEN from evaluating their own system (Q4.4-2).
? Strict Appendices Archiving: All signed permission letters, consent forms, UAT sign-off sheets, raw test case logs, and accomplished ISO/IEC 25010 questionnaires MUST be preserved and bound into the manuscript Appendices.
? Exact Scale Alignment: The 5-point evaluation scale and verbal interpretations reported in Chapter 4 must match Section 3.2.4 verbatim.

2. Master Preparation Summary at a Glance

Requirement Category	Key Deliverables & Action Items	Guide Location
A. Physical Papers to Secure		**Pages 12, 13, 17, 27, 33**
B. Who You Need (3 Groups)		**Pages 29, 32 (Table 4-15, Q4.4-1)**
C. Technical Preparations		**Pages 19, 20, 24, 26 (Table 4-11, 4-12)**
D. Testing & Defect Logs		**Pages 20, 21, 27 (Table 4-13, 4-14)**
E. ISO/IEC 25010 Evaluation		**Pages 28, 29, 30, 31 (Table 4-16 to 4-18)**

3. Complete Breakdown of Required Physical Papers (For Appendices)

1. Site Deployment & Permission Letter
Guide Ref: Page 13, Q4.1-6
Pre-Deployment (Before Mond

2. Informed Consent Forms (RA 10173)
Guide Ref: Page 12, 17, 26 (Q4.2-9)
Pre-Deployment / Day 1

3. Client Records Inventory & Receipt
Guide Ref: Page 15, 17 (Table 4-8, Q4.2-8)
Pre-Deployment

4. Test Cases & Execution Logs
Guide Ref: Page 20, 27 (Table 4-13, Q4.3.8-5)
During Pilot Testing

5. Defect Log & Revision Log
Guide Ref: Page 21, 27 (Table 4-14, Q4.3.8-2)
During & End of Pilot

6. Signed UAT Sign-Off Form
Guide Ref: Page 27, 40 (Q4.3.8-4, Table 14)
End of Pilot Week

7. ISO/IEC 25010 Accomplished Forms
Guide Ref: Page 29, 33 (Table 4-16, Q4.4-11)
Post-Pilot Evaluation Day

8. Station Posters & Quick Guides
Guide Ref: Page 18, 26 (Q4.3.1-4)
Day 1 of Pilot Deployment

4. Stakeholders & Evaluator Matrix (Who You Need)

Evaluator Group	Specific Target Roles	Basis for Inclusion (Guide Table 4-15)	Assigned ISO Charac
Group 1: End Users (Daily Operators)	? Volunteer Tutors ? Subject Teachers ? Attendance/DTR Staff	Daily operators of the system with direct experience in QR attendance, score entry, and risk alerts.	
Group 2: Client Reps (Administrative)	? School Principal ? Head Teacher ? ARAL Coordinator	Accountable for school operations, remedial literacy program policies, and DepEd report generation.	
Group 3: IT Experts (Technical Evaluators)	? External IT Professionals ? CS/IT Faculty Members ? Systems Developers	Able to assess code architecture, password hashing, database design, maintainability, and portability.	

CRITICAL DEFENSE RULE (Guide Page 32, Q4.4-2):
No member of your capstone development group can be an evaluator. Evaluators must be 100% external to the project team.

5. Technical Preconditions & Deployment Environment Setup

A. What You Must Prepare on the System Before Monday (Pages 5, 19, 20, 26):

- **Feature-Complete Code Freeze (Table 2, Q1-4)**
- **Data Sanitation & Privacy Compliance (Part 5.2, Q4.3.4-4)**
- **Account Provisioning & Role-Privilege Matrix (Table 4-11)**
- **Deployment Environment Documentation (Table 4-12, Q4.3.7-1)**
- **Network & Offline Handling Verification (Table 4-1, Q4.3.1-3)**

B. ISO/IEC 25010 Rating Scale & Mathematical Computation (Pages 29·31)

Scale Weight	Range of Means	Verbal Interpretation	Operational Meaning in Capstone Study
5	4.21 to 5.00	**Excellent**	The quality characteristic is fully satisfied with no observed deficie
4	3.41 to 4.20	**Very Satisfactory**	The characteristic is satisfied, with minor room for improvement.
3	2.61 to 3.40	**Satisfactory**	The characteristic is met at an acceptable operational level.
2	1.81 to 2.60	**Fair**	The characteristic is partially met and needs improvement.
1	1.00 to 1.80	**Poor**	The quality characteristic is not met.

Formula for Weighted Mean (WM) per Characteristic (Page 30, Table 4-17):

WM = [· (f × w)] / n where: f = response frequency, w = scale weight (1 to 5), n = total evaluators

Important Arithmetic Rule: All weighted means must be rounded to exactly 2 decimal places and verified before submission.

6. Guide Visual Proof: Preconditions & Permissions

EXHIBIT A: Preconditions for Chapter 4 (Guide Page 5, Table 2)

Highlights that the system must be feature-complete and tested before pilot testing, and evaluation data must use declared scales.



IT 124 | Capstone Project 2 | Technical Documentation and Manuscript Finalization

Table 2. Preconditions for writing Chapter 4

Precondition	Why it matters	Where it comes from
Specific objectives are final, SMART, and lettered	They fix the number of results, the number of conclusions, and the order of both.	Section 1.2 of your manuscript, as approved in Capstone Project 1 and revised after the proposal defense.
Methods are fully described in Chapter 3	Chapter 4 reports outcomes only. Any procedure explained in Chapter 4 is a sign that Chapter 3 is incomplete.	Sections 3.2.1 to 3.2.4.
The system is feature-complete and tested	Section 4.3 reports a finished system, not one still under construction. Screenshots of unfinished modules invite questions you cannot answer.	Outputs of Module 03 on testing and Module 04 on revisions.
Evaluation data has been collected using the declared instrument and scale	The scale reported in Section 4.4 must be the same scale declared in Section 3.2.4. A mismatch is a finding the panel will raise.	Section 3.2.4 and your accomplished evaluation forms. If evaluation happens during the pilot, also Module 06.

Part 1.3 The three-pass working method

Do not start by writing prose. Start by assembling evidence. Groups that write first and look for evidence later produce chapters that cannot be defended.

Table 3. The three passes

Pass	What you do	Output of the pass
Pass 1: Assemble	Collect every artifact you already produced: needs assessment records, survey returns, diagrams, screenshots, test logs, UAT forms, evaluation tallies. Put them in one folder, named by section number.	A complete evidence folder. If a subsection has no evidence, you have found a gap while there is still time to fix it.
Pass 2: Tabulate	Turn raw evidence into the tables and figures that will appear in the chapter. Number them in draft form, such as Table 4-1 and Figure 4-3. Compute all means, totals, and percentages now.	A set of finished exhibits. The chapter is now mostly built, because a good results chapter is a set of exhibits joined by explanation.
Pass 3: Write and audit	Write the narrative around the exhibits, then audit the chapter against your specific objectives using the traceability matrix in Part 7 of this guide.	A draft in which every objective is answered and every exhibit is discussed.

The single rule that governs Chapter 4

Your specific objectives in Section 1.2 are the contract. Chapter 4 reports the evidence that each specific objective was met, in the same order in which the objectives are stated.

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EXHIBIT B: Permissions, Locale Consent & Appendices (Guide Page 13, Q4.1-6 & Q4.1-7)

Directs students to secure institutional consent, obtain deployment permission letters, and archive all interview transcripts and consent forms in Appendices.



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Q4.1-3 Our client had no written records at all. What evidence can we present?

The absence of records is itself a finding, and it can be evidenced. Report the observation, the number of transactions observed over a stated period, the time each step took, and the statements of the personnel who perform the work. A photograph of the physical setup, with personal data masked, may be placed in the Appendices and referred to in the text.

Q4.1-4 We conducted the needs assessment a year ago in Capstone Project 1. Is it still valid?

It is valid if the process has not changed, but you must say so. State the period covered by the assessment and confirm with the client, in writing if possible, that the process remained the same up to development. If something changed, report the change as an update in this section rather than leaving the reader to assume that the old data still holds.

Q4.1-5 Can Figure 4-1 be the same diagram we used in Capstone Project 1?

Yes, if the process it shows is still accurate and the figure is legible at print size. Renumber it to the Chapter 4 scheme, regenerate it at a readable resolution rather than enlarging a small image, and check that the actors and step labels match the terminology used in the rest of Chapter 4.

Q4.1-6 May we name the barangay, the office, or the company?

Naming the institution is normally expected, because the locale is part of the study. Naming individuals is not. Use role labels such as Health Worker 2 or Office Representative. Secure the consent of the institution for the use of its name, and follow whatever the client stipulated in your data gathering permission letter.

Q4.1-7 Where do we put the interview transcripts and accomplished forms?

In the Appendices, referenced by letter from the chapter text. Chapter 4 carries the tabulated findings and one representative statement per theme. A chapter that reproduces whole transcripts is unreadable and is routinely returned for revision.

Q4.1-8 Should the requirements table list every requirement, including the trivial ones?

List the requirements that shaped the system, grouped as functional and non-functional, with a priority column. Very fine-grained requirements can be summarized at the level of the feature they belong to, and the complete list placed in the Appendices. What must not happen is a requirement in the table that the system does not implement.

Q4.1-9 Does Section 4.1 need a figure of the proposed process as well?

The proposed or to-be process belongs in Section 4.3, where the developed system is presented. Keep Section 4.1 to the current situation, so that the contrast between the two is clean and the reader can see exactly what changed.

Q4.1-10 A panelist asked why a certain feature exists. Which exhibit answers that?

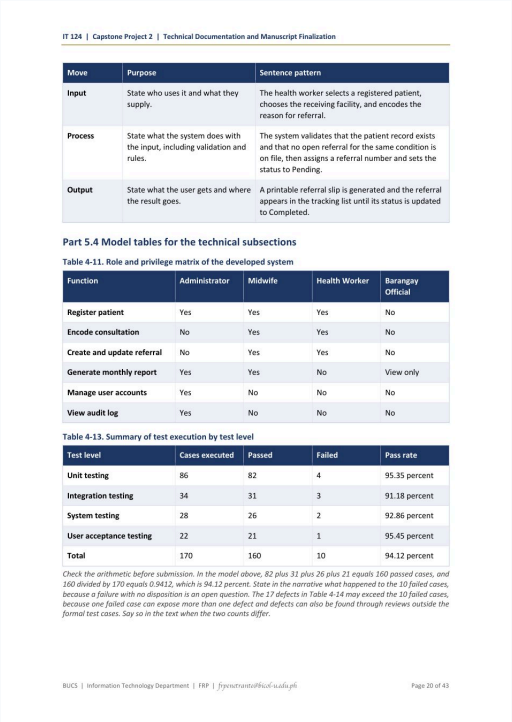
Table 4-3, the requirements traceability table. Reading one row aloud answers the question completely: this problem was found, this requirement was derived from it, and this module implements it. Know the page number of that table before the defense.

BUCS | Information Technology Department | FRP | frpenetrante@bicol-u.edu.ph Page 13 of 43

7. Guide Visual Proof: Roles, Privileges & Deployment Setup

EXHIBIT C: Role & Privilege Matrix and Test Execution Levels (Guide Page 20, Table 4-11 & 4-13)

Model format for mapping user roles to permissions and reporting the 4 test levels (Unit, Integration, System, Acceptance).



IT 124 | Capstone Project 2 | Technical Documentation and Manuscript Finalization

Move	Purpose	Sentence pattern
Input	State who uses it and what they supply.	The health worker selects a registered patient, chooses the receiving facility, and encodes the reason for referral.
Process	State what the system does with the input, including validation and rules.	The system validates that the patient record exists and that no open referral for the same condition is on file, then assigns a referral number and sets the status to Pending.
Output	State what the user gets and where the result goes.	A printable referral slip is generated and the referral appears in the tracking list until its status is updated to Completed.

Part 5.4 Model tables for the technical subsections

Table 4-11. Role and privilege matrix of the developed system

Function	Administrator	Midwife	Health Worker	Barangay Official
Register patient	Yes	Yes	Yes	No
Encode consultation	No	Yes	Yes	No
Create and update referral	No	Yes	Yes	No
Generate monthly report	Yes	Yes	No	View only
Manage user accounts	Yes	No	No	No
View audit log	Yes	No	No	No

Table 4-13. Summary of test execution by test level

Test level	Cases executed	Passed	Failed	Pass rate
Unit testing	86	82	4	95.35 percent
Integration testing	34	31	3	91.18 percent
System testing	28	26	2	92.86 percent
User acceptance testing	22	21	1	95.45 percent
Total	170	160	10	94.12 percent

Check the arithmetic before submission. In the model above, 82 plus 31 plus 26 plus 21 equals 160 passed cases, and 160 divided by 170 equals 0.9412, which is 94.12 percent. State in the narrative what happened to the 10 failed cases, because a failure with no disposition is an open question. The 17 defects in Table 4-14 may exceed the 10 failed cases, because one failed case can expose more than one defect and defects can also be found through reviews outside the formal test cases. Say so in the text when the two counts differ.

BUCS | Information Technology Department | FRP | frpenetrante@bicol-u.edu.ph Page 20 of 43

EXHIBIT D: Deployment Environment Specifications (Guide Page 26, Part 5.13)

Explains that hardware/software specifications must document the client-side testing setup (processors, OS, runtimes, browsers).



IT 124 | Capstone Project 2 | Technical Documentation and Manuscript Finalization

Q4.3.6-3 Our system runs on a local network with no internet exposure. Does security still need a subsection?

Yes. Physical and administrative controls are still controls. Report who has access to the machine, how accounts are provisioned and deactivated, how backups are taken and where they are kept, and what happens to the data if the machine is lost. State the exposure honestly, including any control that is out of your scope.

Q4.3.6-4 How does this subsection relate to the Data Privacy Act of 2012?

The Data Privacy Act of 2012, Republic Act No. 10173, applies whenever your system processes personal information. State what personal data the system holds, the purpose for holding it, who can see it, how long it is retained, and what technical and organizational measures protect it. Reference the Act properly in the text and in the References list.

Q4.3.6-5 Should we report a vulnerability we know about but did not fix?

Report it. Place it as a deferred item in Section 4.3.8 with its severity, and carry it into the recommendations in Section 5.3. A known limitation that is disclosed is a sign of professional judgment. A known limitation that a panelist discovers during the demonstration is a different conversation entirely.

Part 5.13 Frequently asked questions on Section 4.3.7 Deployment Environment

Q4.3.7-1 What belongs in the deployment environment table?

The hardware and software on which the system actually ran during deployment or pilot testing, with versions: processor, memory, and storage of the host machine, operating system, runtime or server software, database engine, and client devices or browsers. Versions matter, because a result obtained on one version is not automatically a result on another.

Q4.3.7-2 Our development machines are far better than the client hardware. Which do we report?

Report both, in separate columns or separate rows, clearly labeled. The deployment environment is the one your performance and reliability claims rest on, so it must be the client-side specification. Reporting only the development machine and then claiming acceptable response time is a mismatch that a panelist can expose in one question.

Q4.3.7-3 We were not able to deploy at the client site. What do we report?

Report what actually happened. If the system was pilot tested in a simulated environment, say so, describe the simulation, state why site deployment did not proceed, and adjust the claims in Sections 4.4 and 5.2 so that they do not assert operational deployment. Then carry the deployment into the recommendations in Section 5.3.

Q4.3.7-4 Do we include the hosting plan, domain, or cloud service details?

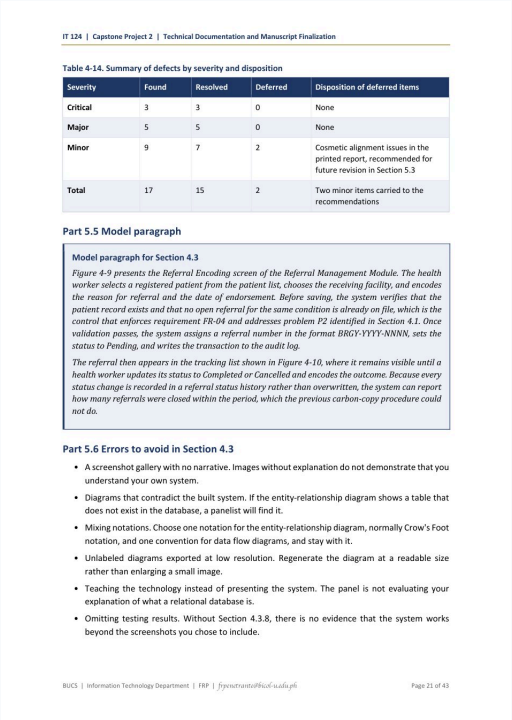
Include the service type and region or tier, because they affect availability and cost, and state who pays for and administers the account after turnover. Do not include account identifiers, keys, or credentials anywhere in the manuscript.

BUCS | Information Technology Department | FRP | frpenetrante@bicol-u.edu.ph Page 26 of 43

8. Guide Visual Proof: Test Execution, Defects & UAT Sign-Off

EXHIBIT E: Defect Severity & Disposition Summary (Guide Page 21, Table 4-14)

Model table showing how found defects must be categorized by severity (Critical, Major, Minor) and resolved before defense.



IT 124 | Capstone Project 2 | Technical Documentation and Manuscript Finalization

Table 4-14. Summary of defects by severity and disposition

Severity	Found	Resolved	Deferred	Disposition of deferred items
Critical	3	3	0	None
Major	5	5	0	None
Minor	9	7	2	Cosmetic alignment issues in the printed report, recommended for future revision in Section 5.3
Total	17	15	2	Two minor items carried to the recommendations

Part 5.5 Model paragraph

Model paragraph for Section 4.3

Figure 4-9 presents the Referral Encoding screen of the Referral Management Module. The health worker selects a registered patient from the patient list, chooses the receiving facility, and encodes the reason for referral and the date of endorsement. Before saving, the system verifies that the patient record exists and that no open referral for the same condition is already on file, which is the control that enforces requirement FR-04 and addresses problem P2 identified in Section 4.1. Once validation passes, the system assigns a referral number in the format BRGY-YYYY-NNNN, sets the status to Pending, and writes the transaction to the audit log.

The referral then appears in the tracking list shown in Figure 4-10, where it remains visible until a health worker updates its status to Completed or Cancelled and encodes the outcome. Because every status change is recorded in a referral status history rather than overwritten, the system can report how many referrals were closed within the period, which the previous carbon-copy procedure could not do.

Part 5.6 Errors to avoid in Section 4.3

- A screenshot gallery with no narrative. Images without explanation do not demonstrate that you understand your own system.
- Diagrams that contradict the built system. If the entity-relationship diagram shows a table that does not exist in the database, a panelist will find it.
- Mixing notations. Choose one notation for the entity-relationship diagram, normally Crow's Foot notation, and one convention for data flow diagrams, and stay with it.
- Unlabeled diagrams exported at low resolution. Regenerate the diagram at a readable size rather than enlarging a small image.
- Teaching the technology instead of presenting the system. The panel is not evaluating your explanation of what a relational database is.
- Omitting testing results. Without Section 4.3.8, there is no evidence that the system works beyond the screenshots you chose to include.

BUCS | Information Technology Department | FRP | frpenetrante@bicol-u.edu.ph Page 21 of 43

EXHIBIT F: User Acceptance Testing (UAT) Sign-off Proof (Guide Page 27, Part 5.14)

Q4.3.8-4 mandates that UAT must be conducted by the intended users/client rep, and the signed UAT form must be kept in Appendices.



IT 124 | Capstone Project 2 | Technical Documentation and Manuscript Finalization

Part 5.14 Frequently asked questions on Section 4.3.8 Testing Results

Q4.3.8-1 What is the difference between the four test levels?

Unit testing exercises individual functions or components in isolation. Integration testing exercises the interactions and interfaces between components, and between the system and any external system it connects to. System testing exercises the complete, integrated system against the requirements. User acceptance testing is conducted by the intended users to confirm that the system is fit for their work. The ISTQB Certified Tester Foundation Level syllabus, version 4.0, names the corresponding levels component testing, component integration testing, system testing, system integration testing, and acceptance testing. Report the levels separately, because a pass rate that hides which level failed is not informative.

Q4.3.8-2 Our pass rate is not 100 percent. Is that a problem?

Not by itself. What matters is the disposition of every failure. State how many failed, what severity they were, which were resolved and retested, and which were deferred with the reason. An unresolved critical defect is a problem. A deferred cosmetic defect, disclosed and carried into the recommendations, is normal engineering practice. For this course, the Module 03 performance standard requires the system to pass at least 90 percent of test cases, with all critical defects identified, logged, and resolved, and Module 04 requires all critical and major defects to be resolved and recorded in a revision log. Place the revision log and the signed UAT sign-off form in the Appendices and refer to them from this subsection.

Q4.3.8-3 How many test cases are enough?

There is no fixed number. Coverage is the standard, not volume. Every functional requirement in Table 4-2 should be exercised by at least one test case, and every business rule in Section 4.3.5 by at least one. Show the mapping if your adviser asks for it, because it converts a raw count into evidence of coverage.

Q4.3.8-4 Who conducts user acceptance testing, and what proof do we present?

The intended users or the client representative conduct it, never the development team. The proof is the signed UAT sign-off form, kept in the Appendices and referenced from this subsection, together with the tabulated results. Do not report UAT results with no corresponding signed form.

Q4.3.8-5 Where do the test cases themselves go?

The full test case document, with steps, expected results, and actual results, belongs in the Appendices. Section 4.3.8 carries the summary tables and the narrative that explains the failures and their disposition.

Q4.3.8-6 Is testing in Section 4.3.8 the same as the evaluation in Section 4.4?

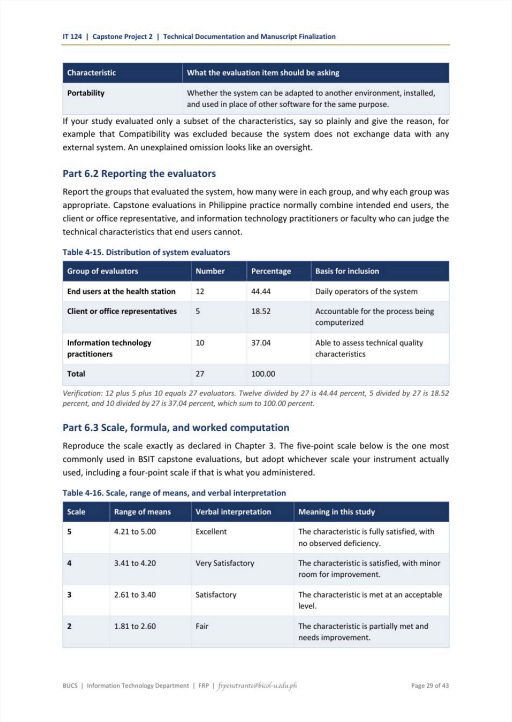
No, and the distinction is frequently confused. Testing asks whether the system behaves as specified, and it is answered by test cases with pass or fail results. Evaluation asks how acceptable the system is to its users and to technical evaluators, and it is answered by rated instruments and weighted means. A system can pass every test case and still be rated poorly on usability.

BUCS | Information Technology Department | FRP | frpenetrante@bicol-u.edu.ph Page 27 of 43

9. Guide Visual Proof: ISO/IEC 25010 Evaluation & Calculations

EXHIBIT G: Evaluator Distribution Table & 5-Point Scale (Guide Page 29, Table 4-15 & 4-16)

Table 4-15 defines the 3 evaluator groups, and Table 4-16 establishes the official scale ranges and verbal interpretations.



IT 124 | Capstone Project 2 | Technical Documentation and Manuscript Finalization

Characteristic	What the evaluation item should be asking
Portability	Whether the system can be adapted to another environment, installed, and used in place of other software for the same purpose.

If your study evaluated only a subset of the characteristics, say so plainly and give the reason, for example that Compatibility was excluded because the system does not exchange data with any external system. An unexplained omission looks like an oversight.

Part 6.2 Reporting the evaluators

Report the groups that evaluated the system, how many were in each group, and why each group was appropriate. Capstone evaluations in Philippine practice normally combine intended end users, the client or office representative, and information technology practitioners or faculty who can judge the technical characteristics that end users cannot.

Table 4-15. Distribution of system evaluators

Group of evaluators	Number	Percentage	Basis for inclusion
End users at the health station	12	44.44	Daily operators of the system
Client or office representatives	5	18.52	Accountable for the process being computerized
Information technology practitioners	10	37.04	Able to assess technical quality characteristics
Total	27	100.00	

Verification: 12 plus 5 plus 10 equals 27 evaluators. Twelve divided by 27 is 44.44 percent, 5 divided by 27 is 18.52 percent, and 10 divided by 27 is 37.04 percent, which sum to 100.00 percent.

Part 6.3 Scale, formula, and worked computation

Reproduce the scale exactly as declared in Chapter 3. The five-point scale below is the one most commonly used in BSIT capstone evaluations, but adopt whichever scale your instrument actually used, including a four-point scale if that is what you administered.

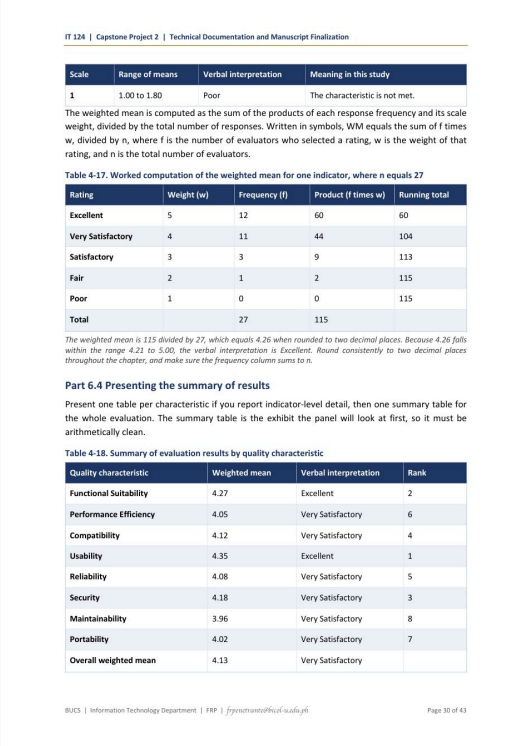
Table 4-16. Scale, range of means, and verbal interpretation

Scale	Range of means	Verbal interpretation	Meaning in this study
5	4.21 to 5.00	Excellent	The characteristic is fully satisfied, with no observed deficiency.
4	3.41 to 4.20	Very Satisfactory	The characteristic is satisfied, with minor room for improvement.
3	2.61 to 3.40	Satisfactory	The characteristic is met at an acceptable level.
2	1.81 to 2.60	Fair	The characteristic is partially met and needs improvement.

BUCS | Information Technology Department | FRP | frpenetrante@bicol-u.edu.ph Page 29 of 43

EXHIBIT H: Worked Weighted Mean Calculation & Summary (Guide Page 30, Table 4-17 & 4-18)

Shows the exact mathematical calculation table for weighted means and the summary table of results across all characteristics.



IT 124 | Capstone Project 2 | Technical Documentation and Manuscript Finalization

Scale	Range of means	Verbal interpretation	Meaning in this study
1	1.00 to 1.80	Poor	The characteristic is not met.

The weighted mean is computed as the sum of the products of each response frequency and its scale weight, divided by the total number of responses. Written in symbols, WM equals the sum of f times w, divided by n, where f is the number of evaluators who selected a rating, w is the weight of that rating, and n is the total number of evaluators.

Table 4-17. Worked computation of the weighted mean for one indicator, where n equals 27

Rating	Weight (w)	Frequency (f)	Product (f times w)	Running total
Excellent	5	12	60	60
Very Satisfactory	4	11	44	104
Satisfactory	3	3	9	113
Fair	2	1	2	115
Poor	1	0	0	115
Total		27	115	

The weighted mean is 115 divided by 27, which equals 4.26 when rounded to two decimal places. Because 4.26 falls within the range 4.21 to 5.00, the verbal interpretation is Excellent. Round consistently to two decimal places throughout the chapter, and make sure the frequency column sums to n.

Part 6.4 Presenting the summary of results

Present one table per characteristic if you report indicator-level detail, then one summary table for the whole evaluation. The summary table is the exhibit the panel will look at first, so it must be arithmetically clean.

Table 4-18. Summary of evaluation results by quality characteristic

Quality characteristic	Weighted mean	Verbal interpretation	Rank
Functional Suitability	4.27	Excellent	2
Performance Efficiency	4.05	Very Satisfactory	6
Compatibility	4.12	Very Satisfactory	4
Usability	4.35	Excellent	1
Reliability	4.08	Very Satisfactory	5
Security	4.18	Very Satisfactory	3
Maintainability	3.96	Very Satisfactory	8
Portability	4.02	Very Satisfactory	7
Overall weighted mean	4.13	Very Satisfactory	

BUCS | Information Technology Department | FRP | frpenetrante@bicol-u.edu.ph Page 30 of 43

10. Guide Visual Proof: Evaluator Rules & Appendices Archiving

EXHIBIT I: Evaluator Eligibility & Standard Edition (Guide Page 32, Q4.4-1 to Q4.4-4)

Explicitly forbids developers from rating their own software (Q4.4-2) and clarifies ISO/IEC 25010 edition consistency.



IT 124 | Capstone Project 2 | Technical Documentation and Manuscript Finalization

- Using a scale in Chapter 4 that does not match the scale declared in Chapter 3, or changing the verbal interpretation labels between chapters.
- Claiming statistical significance, a significant correlation, or a significant difference between groups when no inferential test was run. Descriptive statistics support descriptive claims only.
- Hiding a low score by omitting the characteristic from the summary table.
- Having project members evaluate their own system. Evaluators must be external to the development team.
- Rounding inconsistently, for example reporting 4.1 in one table and 4.13 in another for the same value.

Part 6.7 Frequently asked questions on Section 4.4

Q4.4-1 How many evaluators do we need?

There is no universal number, and your adviser sets the expectation for your project. What the panel checks is whether the composition is defensible: enough actual end users to speak for daily operation, a client representative accountable for the process, and technical evaluators who can judge maintainability, security, and portability. Report the number per group and the basis for including each group, as in Table 4-15.

Q4.4-2 Can members of the group be evaluators?

No. Evaluators must be external to the development team. A group that rates its own system produces a result that cannot be defended, and the question of whether anyone from the group was among the evaluators is asked at almost every defense.

Q4.4-3 Which edition of ISO/IEC 25010 should we use, the 2011 or the 2023?

Use the edition your instrument was built on, and use it in both Chapter 3 and Chapter 4. The 2011 edition has eight characteristics. ISO has withdrawn it, although many existing instruments were built on it. The 2023 edition has nine, adding Safety and replacing Usability with Interaction Capability and Portability with Flexibility. If you are building a new instrument, use the 2023 edition. Mixing the two editions is not defensible, and the panel will ask which one you used.

Q4.4-4 Do we have to evaluate all eight characteristics?

Evaluate the ones that apply, and state plainly why any characteristic was excluded. Compatibility is the one most commonly and most legitimately excluded, when the system exchanges data with no other system. An omission with a stated reason reads as a scope decision. An omission with no explanation reads as an oversight, or worse, as a hidden low score.

Q4.4-5 How do we compute the overall weighted mean?

Take the arithmetic mean of the characteristic means when each characteristic carries equal weight, which is the usual design. Part 6.4 of this guide shows the verification. If your instrument weights characteristics unequally, state the weights in Chapter 3 and show the weighted computation here. Do not switch between methods between tables.

BUCS | Information Technology Department | FRP | frpenetrante@bicol-u.edu.ph Page 32 of 43

EXHIBIT J: Archiving Accomplished Evaluation Forms (Guide Page 33, Q4.4-10 & Q4.4-11)

Mandates that all accomplished evaluation survey sheets and blank instruments must be bound in the Appendices.



IT 124 | Capstone Project 2 | Technical Documentation and Manuscript Finalization

Q4.4-6 Our lowest characteristic scored 3.40, just below the Very Satisfactory range. May we report it as 3.41?

No. Report the computed value to two decimal places and read its verbal interpretation from the declared range, which makes 3.40 Satisfactory. Adjusting a value to move it into a better band is falsification. Instead, discuss the low score honestly, name the probable cause, and carry the fix into the recommendations in Section 5.3.

Q4.4-7 One evaluator gave very low ratings across the board. May we exclude that response?

Only if the response fails a criterion you declared in Chapter 3, such as more than half of the items unanswered or a single rating marked down the whole column with contradictory comments. Declare the exclusion rule before you apply it, report how many responses were excluded and why, and keep the forms in the Appendices. Removing a response because you dislike it is not permissible.

Q4.4-8 May we say that the system is significantly better than the manual process?

Not on the basis of descriptive statistics. Claims such as significant, significantly correlated, and significantly different require an inferential test. If you ran no such test, write that the system was rated Very Satisfactory and that evaluators reported faster retrieval, which is a descriptive claim. If your Chapter 3 declared an inferential test, report the test statistic, the degrees of freedom, the p value, and the decision.

Q4.4-9 Should we present a table for every characteristic, or only the summary?

Present the indicator-level table for each characteristic if your instrument collected indicator-level data, then the summary table. If the chapter becomes unwieldy, place the indicator-level tables in the Appendices and keep the summary in the chapter, but say in the text where the detail can be found. Never present the overall mean alone.

Q4.4-10 The panel asked us to show how a weighted mean was computed. What do we show?

Show a table in the shape of Table 4-17: rating, weight, frequency, product, and total, followed by the division. Know which page that table is on. The reason the question is asked so often is that a group that cannot reproduce its own mean has usually copied the figure from a spreadsheet nobody checked.

Q4.4-11 Where do the accomplished evaluation forms go?

In the Appendices, together with the blank instrument. Chapter 4 carries the tabulated results only. Keep the raw tally sheet as well, because it is what you will use to reproduce a mean if a panelist asks.

Q4.4-12 Is a high evaluation result enough to conclude that the objective was met?

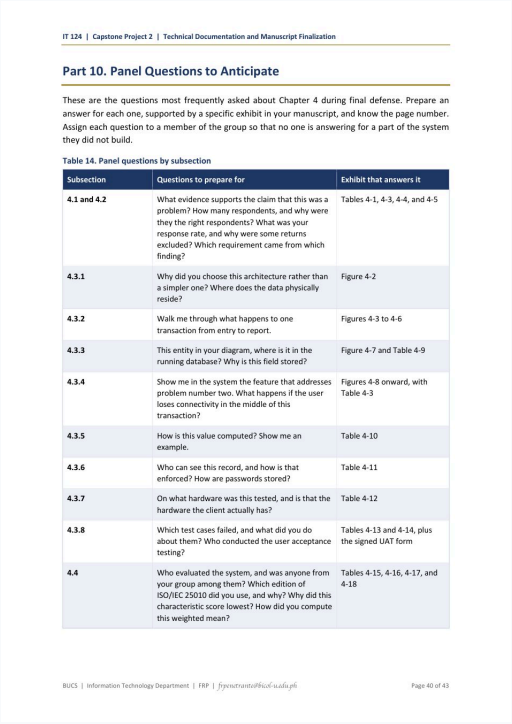
It is the evidence for the evaluation objective only. Each specific objective needs its own evidence. In the four-objective example in Table 11, that means the assessment findings for the first, the developed system and its models for the second, the testing results for the third, and the evaluation for the fourth. A strong Section 4.4 cannot compensate for a missing Section 4.3.8.

BUCS | Information Technology Department | FRP | frpenetrante@bicol-u.edu.ph Page 33 of 43

11. Guide Visual Proof: Panel Defense Questions & Checklist

EXHIBIT K: Panel Defense Questions on Testing & Evaluation (Guide Page 40, Table 14)

Outlines questions panelists ask on defense day regarding UAT sign-offs, hardware environments, evaluators, and means.



IT 124 | Capstone Project 2 | Technical Documentation and Manuscript Finalization

Part 10. Panel Questions to Anticipate

These are the questions most frequently asked about Chapter 4 during final defense. Prepare an answer for each one, supported by a specific exhibit in your manuscript, and know the page number. Assign each question to a member of the group so that no one is answering for a part of the system they did not build.

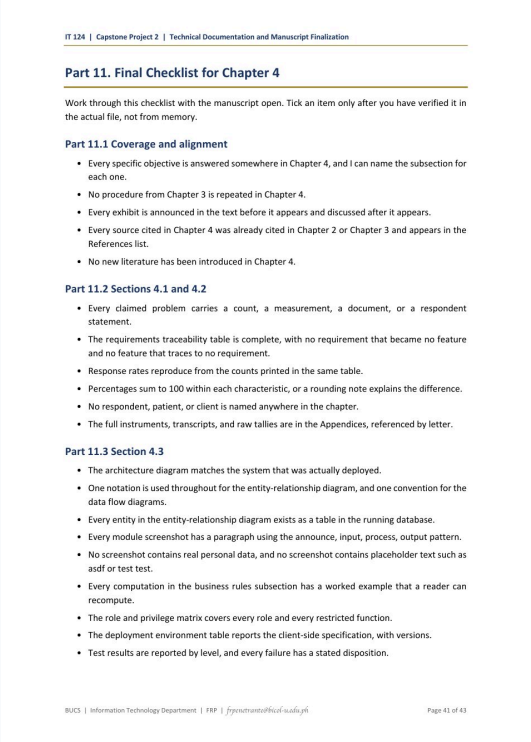
Table 14. Panel questions by subsection

Subsection	Questions to prepare for	Exhibit that answers it
4.1 and 4.2	What evidence supports the claim that this was a problem? How many respondents, and why were they the right respondents? What was your response rate, and why were some returns excluded? Which requirement came from which finding?	Tables 4-1, 4-3, 4-4, and 4-5
4.3.1	Why did you choose this architecture rather than a simpler one? Where does the data physically reside?	Figure 4-2
4.3.2	Walk me through what happens to one transaction from entry to report.	Figures 4-3 to 4-6
4.3.3	This entity in your diagram, where is it in the running database? Why is this field stored?	Figure 4-7 and Table 4-9
4.3.4	Show me in the system the feature that addresses problem number two. What happens if the user loses connectivity in the middle of this transaction?	Figures 4-8 onward, with Table 4-3
4.3.5	How is this value computed? Show me an example.	Table 4-10
4.3.6	Who can see this record, and how is that enforced? How are passwords stored?	Table 4-11
4.3.7	On what hardware was this tested, and is that the hardware the client actually has?	Table 4-12
4.3.8	Which test cases failed, and what did you do about them? Who conducted the user acceptance testing?	Tables 4-13 and 4-14, plus the signed UAT form
4.4	Who evaluated the system, and was anyone from your group among them? Which edition of ISO/IEC 25010 did you use, and why? Why did this characteristic score lowest? How did you compute this weighted mean?	Tables 4-15, 4-16, 4-17, and 4-18

BUCS | Information Technology Department | FRP | fpenetrante@bicol-u.edu.ph Page 40 of 43

EXHIBIT L: Chapter 4 Final Audit Checklist (Guide Page 41, Part 11)

Pre-submission verification checklist ensuring all tables, figures, formulas, and appendices are fully audited.



IT 124 | Capstone Project 2 | Technical Documentation and Manuscript Finalization

Part 11. Final Checklist for Chapter 4

Work through this checklist with the manuscript open. Tick an item only after you have verified it in the actual file, not from memory.

Part 11.1 Coverage and alignment

- Every specific objective is answered somewhere in Chapter 4, and I can name the subsection for each one.
- No procedure from Chapter 3 is repeated in Chapter 4.
- Every exhibit is announced in the text before it appears and discussed after it appears.
- Every source cited in Chapter 4 was already cited in Chapter 2 or Chapter 3 and appears in the References list.
- No new literature has been introduced in Chapter 4.

Part 11.2 Sections 4.1 and 4.2

- Every claimed problem carries a count, a measurement, a document, or a respondent statement.
- The requirements traceability table is complete, with no requirement that became no feature and no feature that traces to no requirement.
- Response rates reproduce from the counts printed in the same table.
- Percentages sum to 100 within each characteristic, or a rounding note explains the difference.
- No respondent, patient, or client is named anywhere in the chapter.
- The full instruments, transcripts, and raw tallies are in the Appendices, referenced by letter.

Part 11.3 Section 4.3

- The architecture diagram matches the system that was actually deployed.
- One notation is used throughout for the entity-relationship diagram, and one convention for the data flow diagrams.
- Every entity in the entity-relationship diagram exists as a table in the running database.
- Every module screenshot has a paragraph using the announce, input, process, output pattern.
- No screenshot contains real personal data, and no screenshot contains placeholder text such as asdf or test test.
- Every computation in the business rules subsection has a worked example that a reader can recompute.
- The role and privilege matrix covers every role and every restricted function.
- The deployment environment table reports the client-side specification, with versions.
- Test results are reported by level, and every failure has a stated disposition.

BUCS | Information Technology Department | FRP | fpenetrante@bicol-u.edu.ph Page 41 of 43